

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 1B- Basic Concepts in Probability 2

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1 Conditional Probability and Bayes' Rule

- Conditional Probability
- Independent Events
- Total Probability Rule
- Bayes' Theorem
- Applications of Bayes' Theorem and TPR

Session Overview

Probability plays a major role in everyday life and decision making depends on one or more events of uncertainty.

In this session, students will be introduced to conditional events and probabilities, independent events and probability, total probability rule and Bayes Theorem.

Learning Outcomes

At the end of the session, the student will:

- 1 Understand dependent and independent events.
- 2 Calculate conditional probabilities of random events.
- 3 Apply total probability rule and Bayes theorem for solving real life problem.

Conditional Probability

- Suppose S is the sample space for an experiment with events A and B .
- Then for experiments with equally likely outcomes, the conditional probability of A given B is the fraction of outcomes in B that are also in A .
- Since only outcomes in B are taken into consideration, in effect, B becomes the sample space, so we have:

$$P(A|B) = \frac{n(\text{event } A \text{ occurs given that event } B \text{ occurs})}{n(\text{event } B \text{ occurs})}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ where } P(B) > 0$$

Conditional Probability Cont'd.

- Similarly, we can define $P(B|A)$ as:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}, \text{ where } P(A) > 0.$$

- Also a very useful formula in computing the joint probability of events is

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

- In general, for events $A_i (i = 1, 2, 3, \dots)$

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n)$$

$$= P(A_1)P(A_2|A_1)P(A_3|A_1, A_2)P(A_4|A_1, A_2, A_3) \dots P(A_n|A_1, A_2, \dots, A_{n-1})$$

- The above formula is known as **Multiplication Rule**.

Example 1

A survey of attitudes about one's job yields the data in the following table

Jobs	Happy	Unhappy	Total
Bus Drivers	50	75	125
Lawyers	40	35	75
Total	90	110	200

A person from this group is selected at random. Given that the selected person is a bus driver, find the probability that he or she is happy.

Example Cont'd: Solution

Let H denote the event that a person is happy and B , the event that a bus driver is selected. The required probability is $P(H|B)$

Now, using the conditional probability formula we have:

$$P(H|B) = \frac{P(H \cap B)}{P(B)}$$

From the table, $P(H \cap B) = \frac{50}{200}$ and $P(B) = \frac{125}{200}$

Therefore,

$$P(H|B) = \frac{50/200}{125/200} = \frac{50}{125} = 0.4$$

Example 2

There are 8 boxes, 3 of type A and 5 of type B . Type A boxes each contain 3 green marble and 4 red marbles. Type B boxes each contain 2 green marble and 4 red marbles. A box is selected at random and a marble is drawn at random from the selected box.

- i Find the probability that a red marble is selected.
- ii A marble was selected and was found to be red, what is the probability that it came from a type A box.

Solution:

Let R denote the event that a red marble is selected, let A represent the event that a type A box is selected, and let B denote the event that a type B box is selected.

- ① The required probability is $P(R)$.

Note: a red marble may either come from box A or box B but not both, hence $R = (R \cap A) \cup (R \cap B)$, since $(R \cap A)$ and $(R \cap B)$ are mutually exclusive events,

$$\implies P(R) = P(R \cap A) + P(R \cap B)$$

Using the multiplication formula, we have:

$$P(R \cap A) = P(R|A)P(A)$$

Given that 3 of the 8 boxes are of type A , $P(A) = \frac{3}{8}$.

Next, type A box contains 7 marbles, of which 4 are red.

① Continued:

Therefore, the conditional probability is; $P(R|A) = \frac{4}{7}$.

By the multiplicative rule,

$$P(R \cap A) = P(R|A)P(A) = \frac{4}{7} \times \frac{3}{8} = \frac{3}{14}.$$

However, $P(B) = \frac{5}{8}$ and $P(R|B) = \frac{4}{6} = \frac{2}{3}$ and multiplicative rule yield the result;

$$P(R \cap B) = P(R|B)P(B) = \frac{2}{3} \times \frac{5}{8} = \frac{5}{12}.$$

$$\therefore P(R) = P(R \cap A) + P(R \cap B) = \frac{3}{14} + \frac{5}{12} = 0.63$$

- ii The required probability is $P(A|R)$.
Using the conditional probability formula, we have:

$$P(A|R) = \frac{P(A \cap R)}{P(R)} = \frac{0.21}{0.63} = 0.33$$

From (i) above.

Independents Events

- Two events, A and B that are related in such a way that the occurrence of either one has no bearing on the occurrence of the other are said to be independent.
- For instance, if a coin is tossed repeatedly, the result of one toss does not affect the result of any other toss.
- Alternatively; Two events A and B are said to be independent if knowledge of the occurrence of A , in no ways influences the probability of the occurrence of B and vice versa.
- Statistically, events A and B are said to be independent, if and only if

$$P(A \cap B) = P(A) \times P(B).$$

Independent Events Cont'd.

- It follows from the definition of conditional probability that if A and B are independent, then

$$P(A|B) = P(A) \quad \text{and} \quad P(B|A) = P(B).$$

- We say that three events A, B, and C are jointly or mutually independent if and only if the following conditions hold:

- 1 $P(A \cap B) = P(A) \times P(B)$
- 2 $P(A \cap C) = P(A) \times P(C)$
- 3 $P(B \cap C) = P(B) \times P(C)$
- 4 $P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$

Example 3

A ball is drawn at random from an open box containing four balls numbered 1, 2, 3, 4. Let $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{1, 4\}$.

Examine whether or not A, B, C are independent.

Solution:

For events A, B, C to be independent then, the conditions stated above must hold

$$A \cap B = \{1\} \implies P(A \cap B) = \frac{1}{4}$$

$$A \cap C = \{1\} \implies P(A \cap C) = \frac{1}{4}$$

$$B \cap C = \{1\} \implies P(B \cap C) = \frac{1}{4}$$

$$\textcircled{1} P(A \cap B) = P(A) \times P(B) = \frac{2}{4} \times \frac{2}{4} = \frac{1}{4}$$

$$\textcircled{2} P(A \cap C) = P(A) \times P(C) = \frac{2}{4} \times \frac{2}{4} = \frac{1}{4}$$

$$\textcircled{3} P(B \cap C) = P(B) \times P(C) = \frac{2}{4} \times \frac{2}{4} = \frac{1}{4}$$

$$\text{Also; } A \cap B \cap C = \{1\} \implies P(A \cap B \cap C) = \frac{1}{4}$$

$$\text{But } P(A) \times P(B) \times P(C) = \frac{2}{4} \cdot \frac{2}{4} \cdot \frac{2}{4} = \frac{1}{8} \neq P(A \cap B \cap C).$$

Hence A, B, C are pairwise independent but not jointly or mutually independent.

Theorem 3:

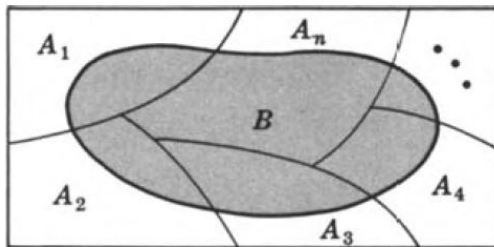
If A and B are independent events, then

- i. A' and B' are independent.
- ii. A' and B are independent.
- iii. A and B' are independent.

The proofs are left as an Exercise.

Total Probability Rule

- Suppose that the sample space, S of a random experiment, can be partitioned into a set of n mutually exclusive and exhaustive events, say $A_1, A_2, \dots, A_n$ (Figure 1).
- Let B be another event in the sample space with a strictly positive probability of taking place.



B is shaded.

Figure: 1

Total Probability Rule Cont'd

- Then

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i), \text{ where } P(A_i) > 0.$$

- **Sketch of the Proof:**
- From Figure 1 above, we can write

$$B = (B \cap A_1) \cup (B \cap A_2) \cup (B \cap A_3) \cup \cdots \cup (B \cap A_n)$$

$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3) + \cdots + P(B \cap A_n)$$

$$\Rightarrow P(B) = \sum_{i=1}^n P(B \cap A_i)$$

Total Probability Rule (Cont..)

- Since, $(B \cap A_i)$, where $i = 1, 2, \dots, k$ are mutually exclusive events.
- And the;

$$P(B|A_i) = \frac{P(B \cap A_i)}{P(A_i)}, \text{ where } P(A_i) > 0$$

- $\implies P(B \cap A_i) = P(B|A_i)P(A_i)$, where $i = 1, 2, \dots, n$

- Therefore

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i) \quad (1)$$

Bayes' Theorem

- Suppose that the events $A_1, A_2, \dots, A_n$, is a partition of S , sample space and B is any event. Then for any i ,

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \dots + P(A_n)P(B|A_n)}$$

- Sketch Proof:**

- Now,

$$P(A_i|B) = \frac{P(A_i \cap B)}{P(B)}, \text{ where } P(B) > 0$$

- Also, $P(B|A_i) = \frac{P(B \cap A_i)}{P(A_i)}$, where $P(A_i) > 0$

$$\Rightarrow P(B \cap A_i) = P(B|A_i)P(A_i).$$

Bayes' Theorem Cont'd.

- Hence,

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{i=1}^n P(B|A_i)P(A_i)} \quad (2)$$

- Equations (1) and (2) are known as Total Probability Rule and Bayes' Theorem, respectively.

Example 1:

A company producing electric relays has three manufacturing plants producing 50, 30 and 20 percent, respectively of its products. Suppose that the probabilities that a relay manufactured by these plants is defective are 0.02, 0.05 and 0.01, respectively.

- a. If a relay is selected at random from the output of the company, what is the probability that it is defective?
- b. If a relay selected at random is found to be defective, what is the probability that it was manufactured by plant 2?

- (a) Let B be the event that the relay is defective, and let A_i be the event that the relay is manufactured by plant i , where $(i = 1, 2, 3)$. The desired probability is $P(B)$.

- Therefore, we have

$$P(B) = \sum_{i=1}^3 P(B|A_i)P(A_i)$$

$$P(B) = 0.02(0.5) + (0.05)(0.3) + (0.01)(0.2) = 0.027$$

- (b) The desired probability is

$$P(A_2|B) = \frac{P(B|A_2)P(A_2)}{P(B)} = \frac{(0.05)(0.3)}{0.027} = 0.556$$

Example 2:

It is estimated that 50% of emails are spam emails. Some software has been applied to filter these spam emails before they reach your inbox. A certain brand of software claims that it can detect 99% of spam emails, and the probability for a false positive (a non-spam email detected as spam) is 5%.

Now if an email is detected as spam, then what is the probability that it is in fact a non-spam email?

Solution:

Let define the following events:

A = event that an email is detected as spam,

B = event that an email is spam,

$\bar{B}$ = event that an email is not spam.

Given that, $P(B) = P(\bar{B}) = 0.5$, $P(A|B) = 0.99$, $P(A|\bar{B}) = 0.05$

Using Bayes' theorem or formula, we have:

$$\begin{aligned} P(\bar{B}|A) &= \frac{P(A|\bar{B})P(\bar{B})}{P(A|B)P(B) + P(A|\bar{B})P(\bar{B})} \\ &= \frac{0.05 \times 0.5}{0.99 \times 0.5 + 0.05 \times 0.5} = 0.048 \end{aligned}$$

Example 3:

Suppose we have 3 cards identical in form except that both sides of the first card are coloured red, both sides of the second card are coloured black, and one side of the third card is coloured red and the other side is coloured black. The 3 cards are mixed up in a hat, and 1 card is randomly selected and put down on the ground.

If the upper side of the chosen card is coloured red, what is the probability that the other side is coloured black?

Solution:

Let RR, BB, and RB denote, respectively, the events that the chosen card is the red-red, the black-black, or the red-black card. Letting R be the event that the upturned side of the chosen card is red, we have that the desired probability is obtained by

$$P(RB|R) = \frac{P(RB \cap R)}{P(R)}$$

$$\begin{aligned} P(RB|R) &= \frac{P(R|RB)P(RB)}{P(R|RB)P(RB) + P(R|RR)P(RR) + P(R|BB)P(BB)} \\ &= \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{1}{3}. \end{aligned}$$

Note that there were 3 black sides and 3 red sides, yielding the $P(RB) = \frac{1}{3} = P(RR) = P(BB)$

Example 4:

In certain juice production company there are three production machines involved in the production. Suppose that machine M_1 produces 30% of the day's production of the product, while machines M_2 and M_2 produce 25% and 45%, respectively. Suppose further that, machines M_1 , M_2 and M_3 turn out 2%, 3% and 4% defective products, respectively, each day. If an item is picked at random and found to be defective, what is the probability that it did not come from machine M_1 ?

Solution:

Let M_1, M_2, M_3 denote the sets of products produced by three machines of the company, respectively. Let D denote the event that an item is defective. The required probability is $P[(M_1|D)']$.

Now, we have $P(M_1) = 0.30$, $P(M_2) = 0.25$, $P(M_3) = 0.45$,
 $P(D|M_1) = 0.02$, $P(D|M_2) = 0.03$ and $P(D|M_3) = 0.04$.

By Bayes' formula, we have

$$\begin{aligned}P(M_1|D) &= \frac{P(D|M_1)P(M_1)}{P(D|M_1)P(M_1) + P(D|M_2)P(M_2) + P(D|M_3)P(M_3)} \\&= \frac{(0.02)(0.30)}{(0.02)(0.30) + (0.03)(0.25) + (0.04)(0.45)} \\&= 0.19\end{aligned}$$

$$\therefore P[(M_1|D)'] = 1 - P(M_1|D) = 0.81$$

Trial Questions:

- 1 In a study, physicians were asked what the odds of breast cancer would be in a woman who was initially thought to have a 1% risk of cancer but who ended up with a positive mammogram result (a mammogram accurately classifies about 80% of cancerous tumors and 90% of benign tumors). Ninety-five (95) out of a hundred physicians estimated the probability of cancer to be about 75%. Do you agree?

Trial Questions:

- 2 Three manufactures M_1 , M_2 and M_3 , supply all of the electronic components to a manufacturer of VCR's. Each VCR uses one unit of the components. Suppose M_1 supplies 41 units, 15% of which are defective; that M_2 supplies 63 units, 10% of which are defective; and M_3 supplies 54 units, 12% of which are defective. A VCR is selected at random and is found to be defective. Which manufacturer is most likely to have provided the components for the defective VCR? Explain your answer.

THANK YOU!!!